SynPhNe™ Enabling you for life

PERFORMANCE REPORT

NAME:	[redacted]	**DATE OF SESSION:**	16/Feb/22
AGE / GENDER:	63-year-old, Male	**REPORTED CONDITION:**	Left MCA infarct
OCCUPATION:	Business	**DURATION OF CONDITION:**	1 year
ADDITIONAL INFO:			

PERFORMANCE SUMMARY

BODY STATE (EMG)		
DURING MOVEMENT CALIBRATION	PRONATOR & SUPINATOR	Supinator muscle shows no isolated agonist contraction, but antagonist contraction is seen throughout. Pronator muscle shows some isolated agonist contraction in between but maximum antagonist muscle activity is seen.
	THUMB	Thumb muscle shows no agonist contraction, and antagonist contraction throughout.
	WRIST EXTENSOR & WRIST FLEXOR	Wrist extensor shows good, isolated agonist contraction throughout with less relaxation however, some antagonist contraction is seen in between. Wrist flexor shows no agonist contraction but only antagonist activity is seen throughout.
	FINGER EXTENSOR & FINGER FLEXOR	Finger extensor shows good agonist contraction with some relaxation, and minimum antagonist activity is seen. Finger flexor shows no agonist activation, but antagonist contraction is seen throughout.
DURING TRAINING & ACTIVITIES	WRIST EXTENSION & FINGERS EXTENSION	Both Wrist extensor and finger extensor show good, isolated agonist contraction but no relaxation at all.
	PICKING UP A PEN & GRASPING A BOTTLE	During PUP and GAB, no agonist contraction is seen, but only antagonist muscle activity is seen throughout the task. This indicates that he is compensating with the stronger muscles while performing the tasks.

BRAIN STATE (EEG)		
DURING THIS SESSION	BRAIN SYMMETRY INDEX	Pre and post session analysis show negative value indicating that she is using her right hemisphere of the brain while performing tasks with the right hand. This indicates non-affected right hemisphere compensation.
	ATTENTION RESPONSE INDEX	Attention response index is 2.902 (>1.1) indicating that he was unable to maintain the attention and focus during the session.
	SMILEY INDEX	Smiley index is 42.9% which indicates that he is not able to maintain the ideal learning state throughout session and hence learning is not optimum.

RECOMMENDATION

1) Regular training with the SynPhNe technology for improving self-regulation to correct muscle use strategy, strengthen muscle and then improve chances of recovery of function.
2) To improve attention and focus during movement and activities by building endurance and maintaining relaxation/attention while in movement (dynamic relaxation training) and performing independence tasks such as holding objects and using right hand during bimanual activities.
3) Improve the relaxation of all the upper limb muscles.
4) To work towards improving Gross motor and Fine Motor Activities by aiming to achieve the individual muscle movement.

If you would like to discuss any of the above, please reach out to us:

SynPhNe Care Centre
Clinic No. 409, 4th Floor, Doctor House
Pedder Road, Cumballa Hill, Mumbai
Maharashtra 400026
+91 8169571552

Opening Hours:
Monday to Saturday 0900 – 1800
Closed on Sunday and Public Holidays

Name: Dr. Ankita Mehendale
Email: ankita@synphne.com

BODY STATE (EMG) INTRODUCTION

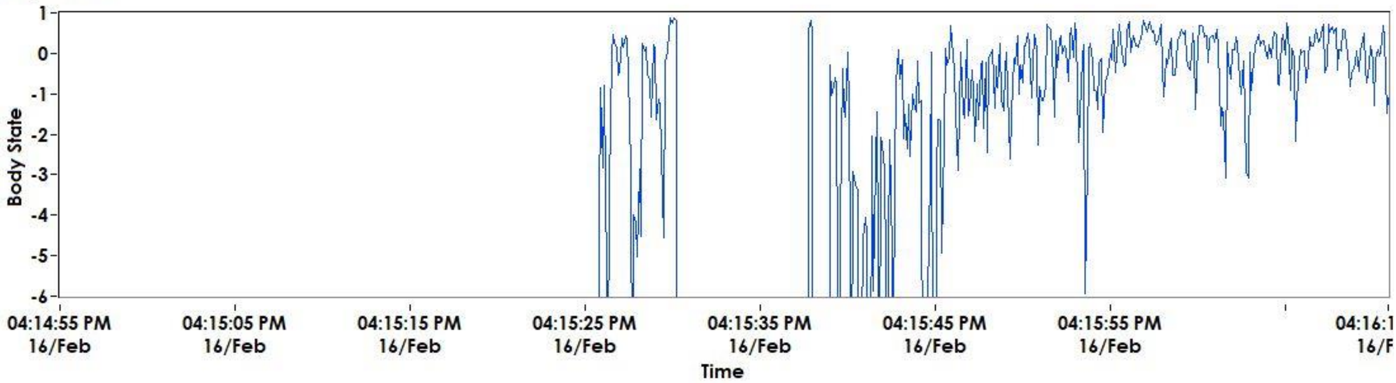
SynPhNe™ Body State (EMG) graphs display the user's primary agonist and antagonist muscle activity read in real-time while the user is attempting to do an upper limb movement, exercise or task, and reflects the quality of agonist-antagonist control.

Reference Range:

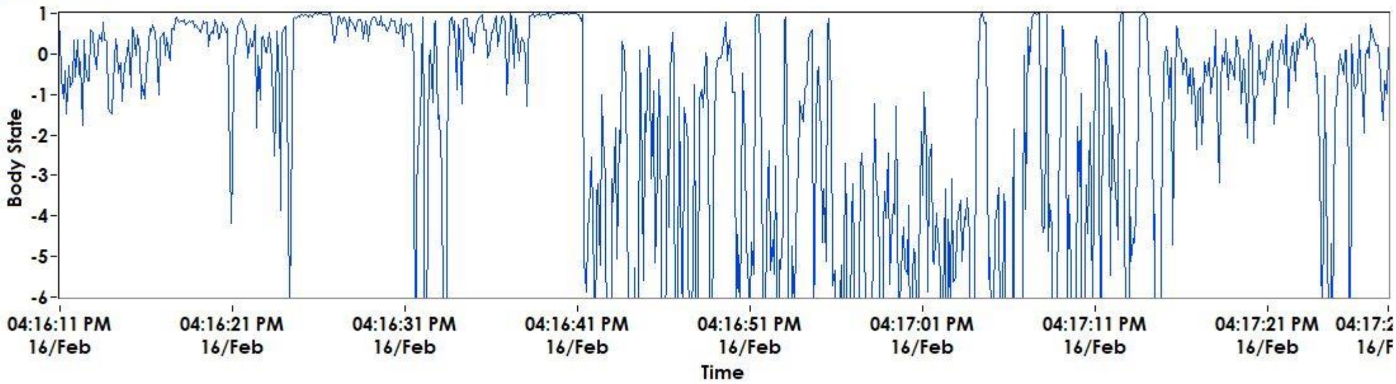
< 0	:	Antagonist muscle is activated more than agonist muscle
= 0	:	Ideal agonist-antagonist relaxed state
> 0	:	Agonist muscle is activated more than antagonist muscle
= 1	:	Ideal agonist contracted state

BODY STATE (EMG) DURING MOVEMENT CALIBRATION

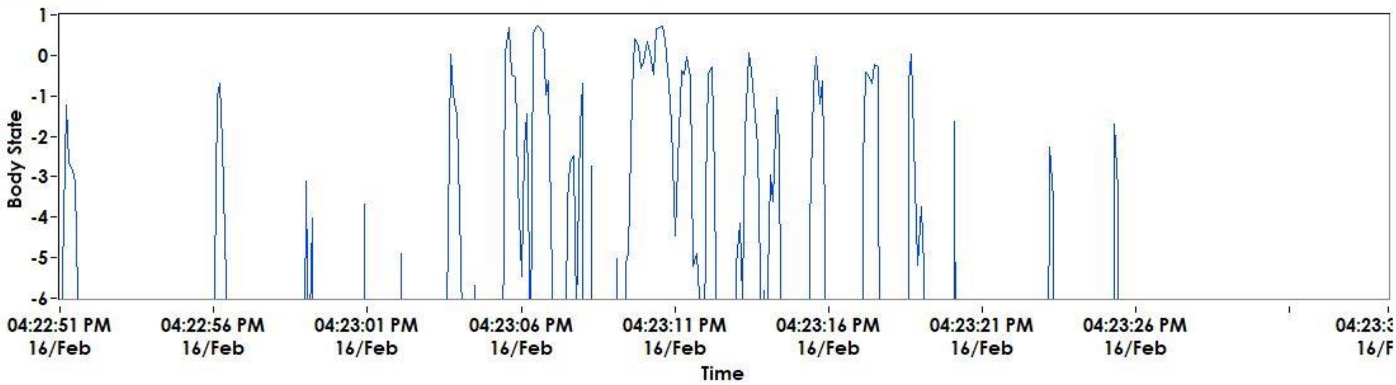
Supinator



Pronator

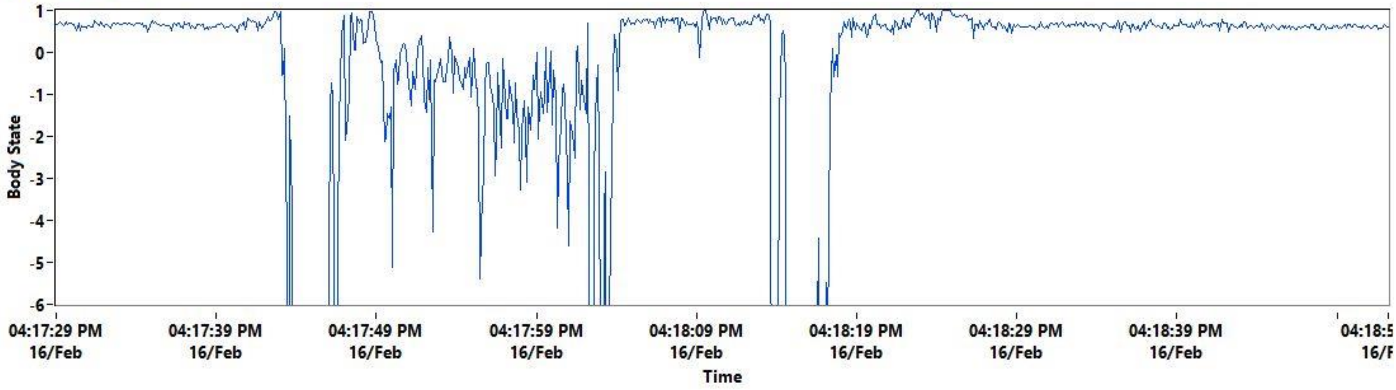


Thumb

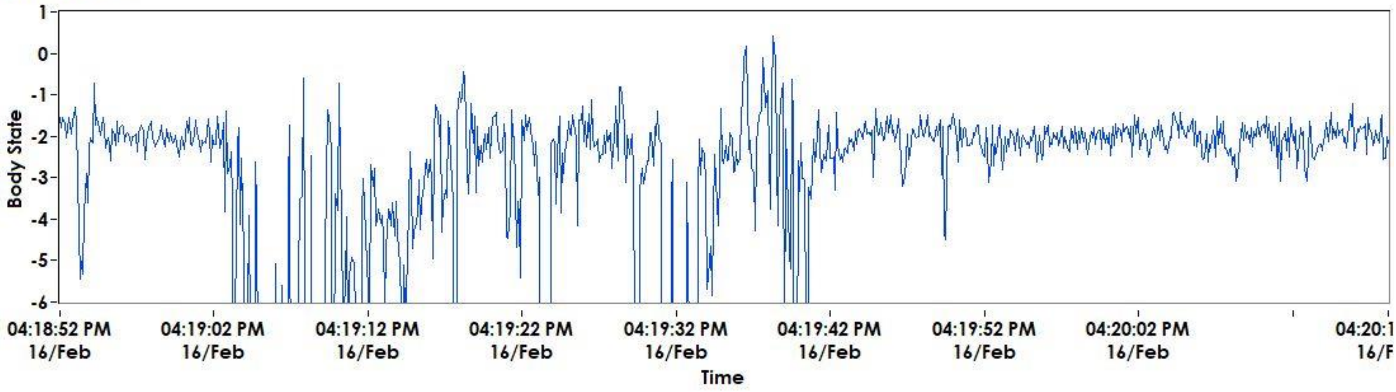


PERFORMANCE REPORT

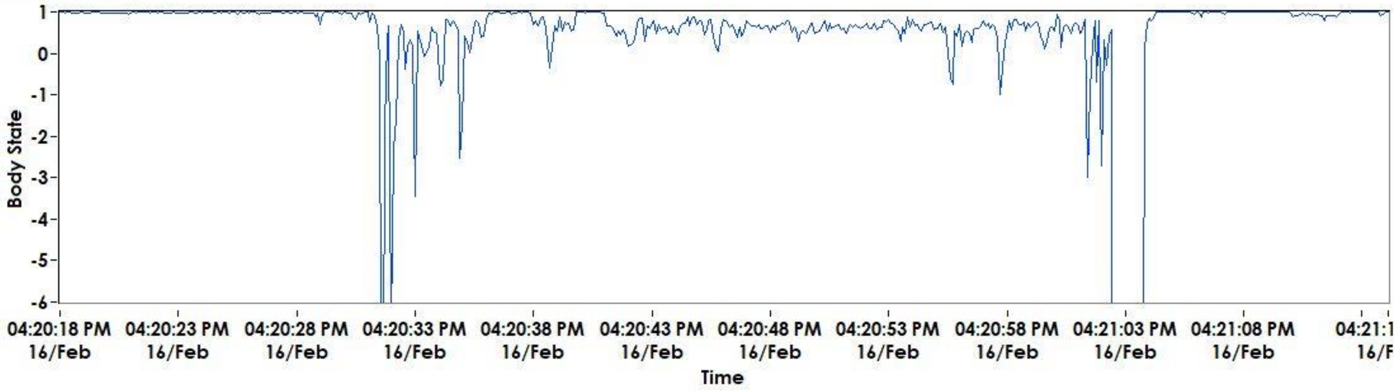
WristExtensor



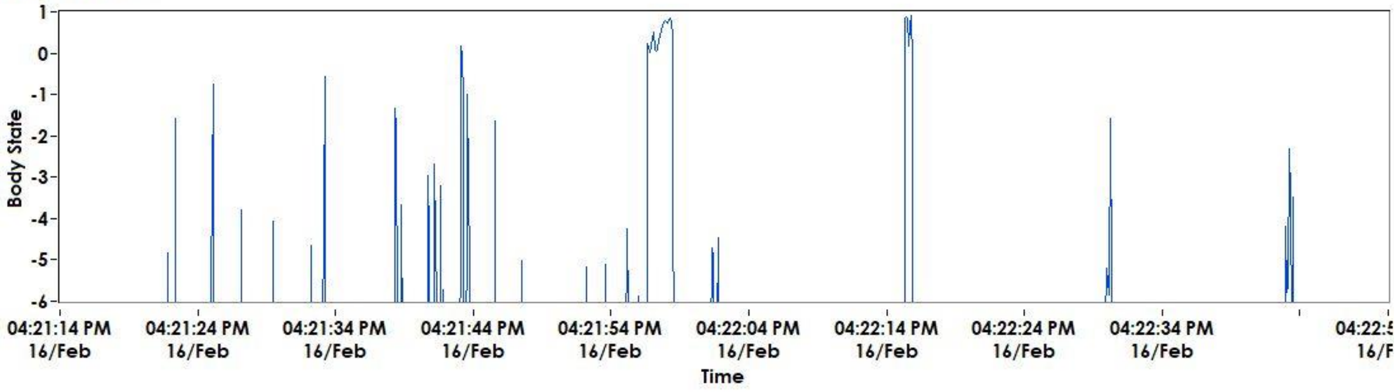
WristFlexor



FingerExtensor

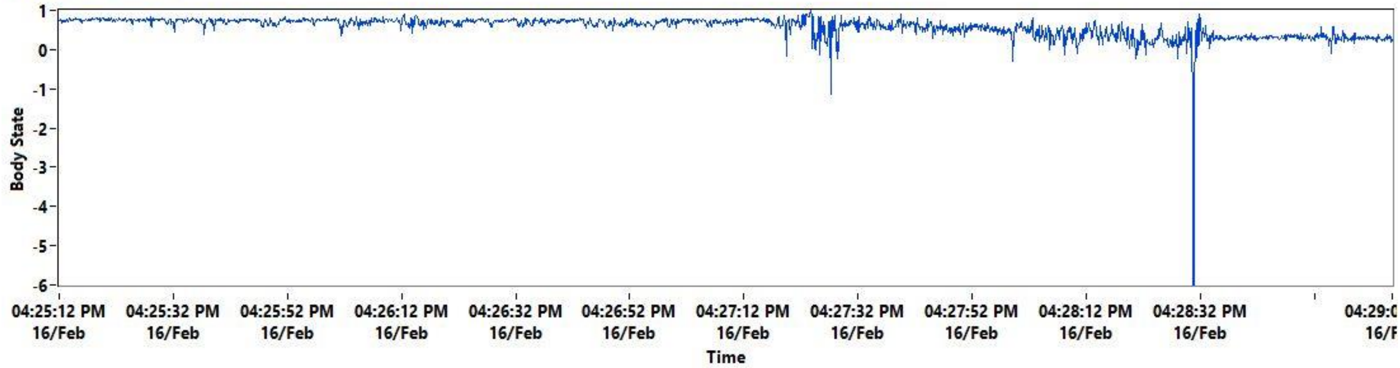


FingerFlexor

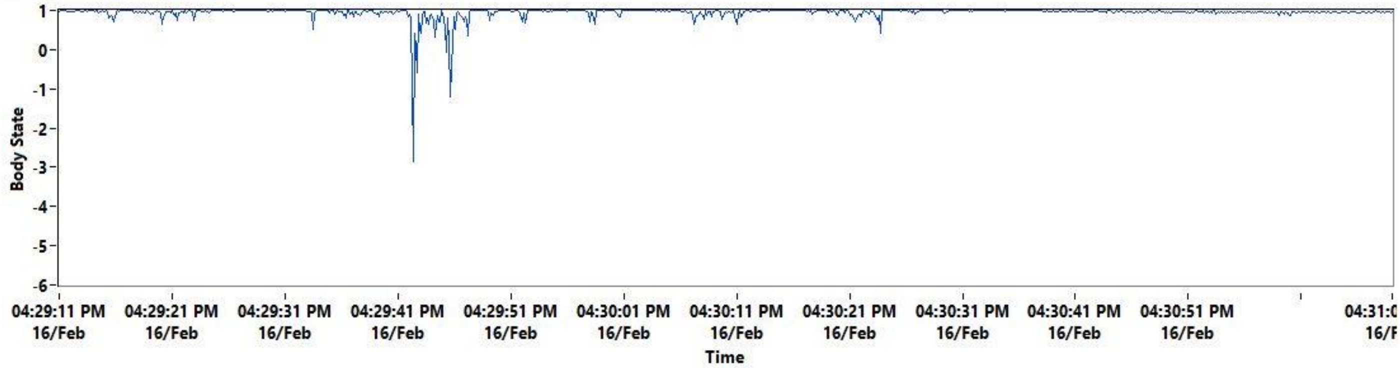


BODY STATE (EMG) DURING AGONIST-ANTAGONIST TRAINING

Wrist Extension

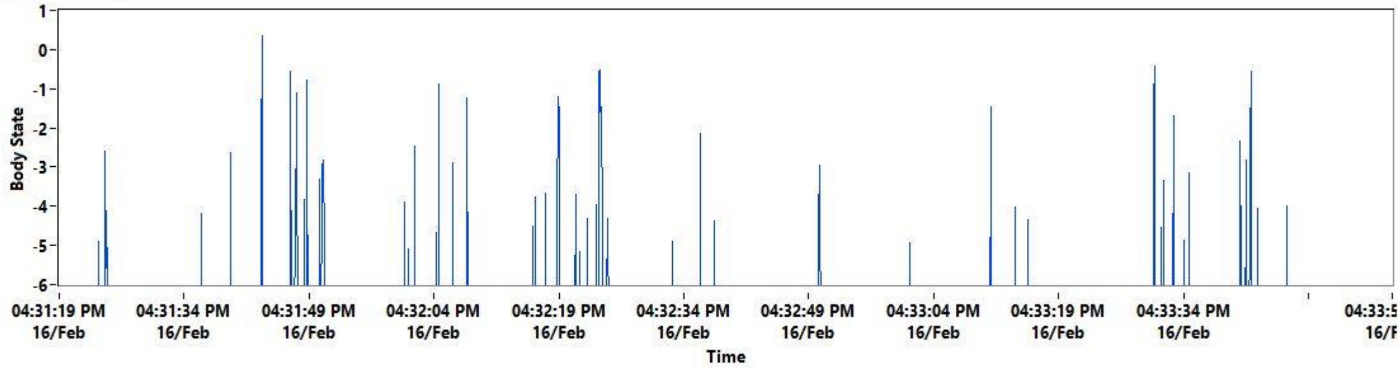


Fingers Extension

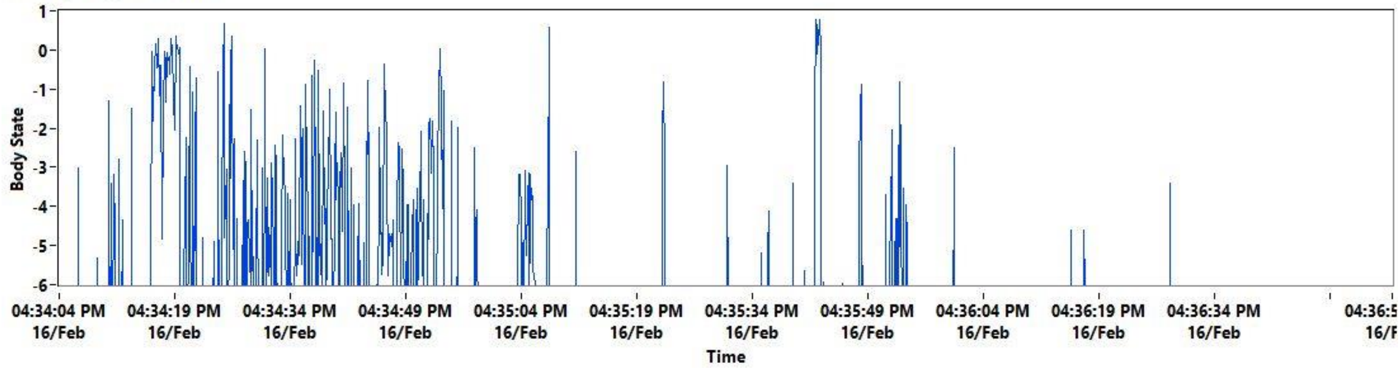


BODY STATE (EMG) DURING ACTIVITIES OF DAILY LIVING

Picking up a Pen



Grasping a Bottle



BRAIN STATE (EEG) INTRODUCTION

SynPhNe™ Brain State (EEG), in this report, focuses on Brain Symmetry Index, Attention Response Index, Smiley Index, and Brain Map which is derived from processing four different wavelengths – alpha, beta, theta and delta – read in real-time while the user is attempting to do a movement, exercise or task.

These indicate the user's unconscious brain response to an upper limb movement, exercise or task.

BRAIN SYMMETRY INDEX

SynPhNe™ Brain Symmetry Index is the difference in EEG-quantified interhemispheric power.

Before Session	After Session
-0.193649	-0.840972

Reference Range

> 0	:	Left Hemisphere is more active than the Right Hemisphere; Left Asymmetrical
= 0	:	Both left & right hemisphere are equally active; Symmetrical
< 0	:	Right hemisphere is more active than the Left Hemisphere; Right Asymmetrical

ATTENTION RESPONSE INDEX

SynPhNe™ Attention Response Index is the difference in EEG-quantified pre-post attention metrics.

Attention Response
2.902

Reference Range

> 1.1	:	Poor attention response
0.9 - 1.1	:	No change in pre-post attention response
< 0.9	:	Good attention response

SMILEY INDEX

SynPhNe™ Smiley Index is a measure of a relaxed and attentive learning brain state. The ideal learning state is achieved if the user is able to maintain the ideal smiley face for more than 70% of the entire session. In this state, learning is optimum and at a relatively low energy cost.

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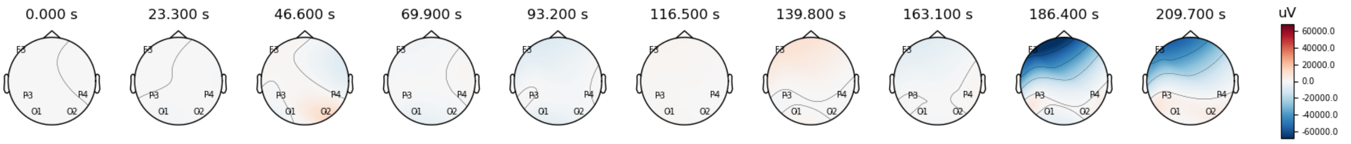
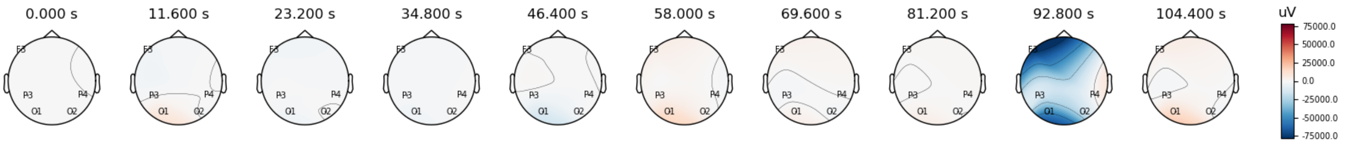
BRAIN MAP (EEG) DURING AGONIST-ANTAGONIST TRAINING

Wrist Extension

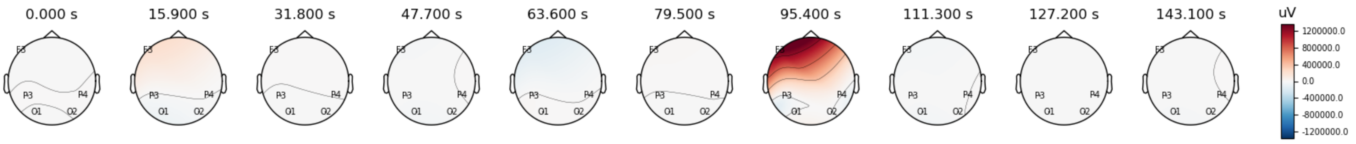


Fingers Extension

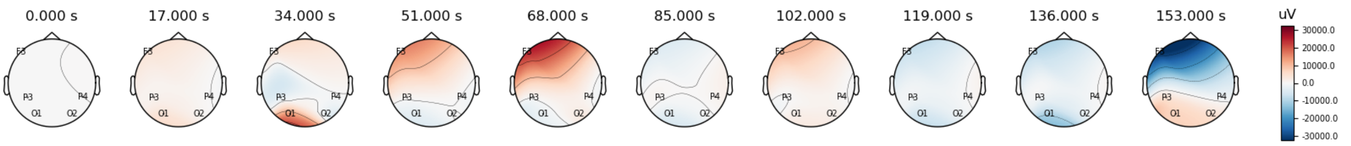


BRAIN MAP (EEG) DURING ACTIVITIES OF DAILY LIVING

Picking up a Pen



Grasping a Bottle



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User's Progress Report

USER:	[redacted]
CONDITION:	MCA stroke
DATE (FROM / TO):	24/02/2022 TO: 24/03/2022
SESSION NO. FROM:	1 TO: 12
NO. OF SESSIONS:	12
DATE OF REPORT:	29 Mar 2022
NAME OF TRAINER:	Dr. Ankita Mehendale

OVERALL SUMMARY

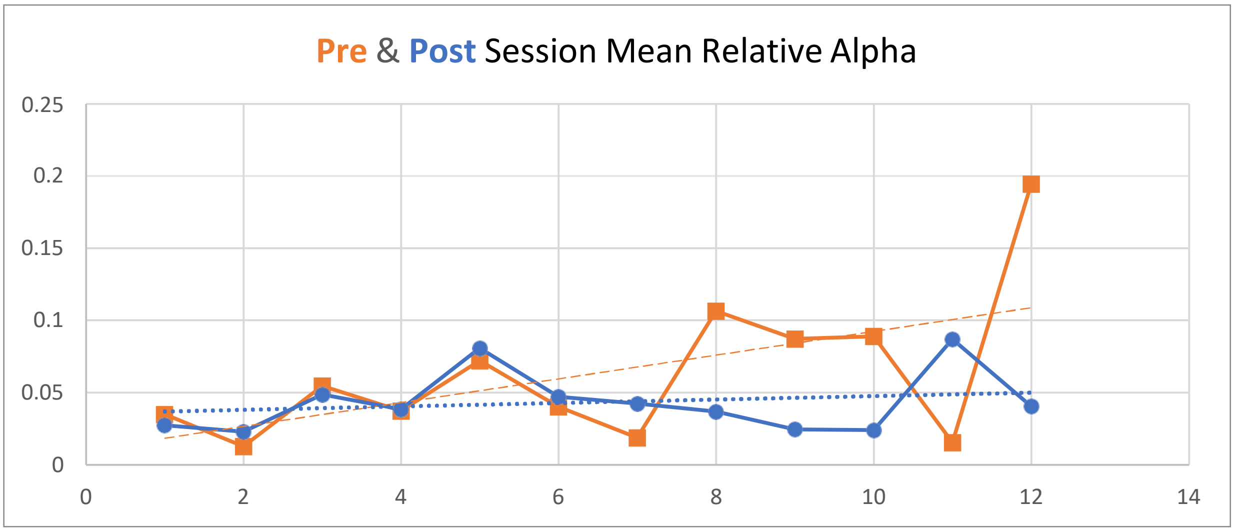
OBSERVATIONS:

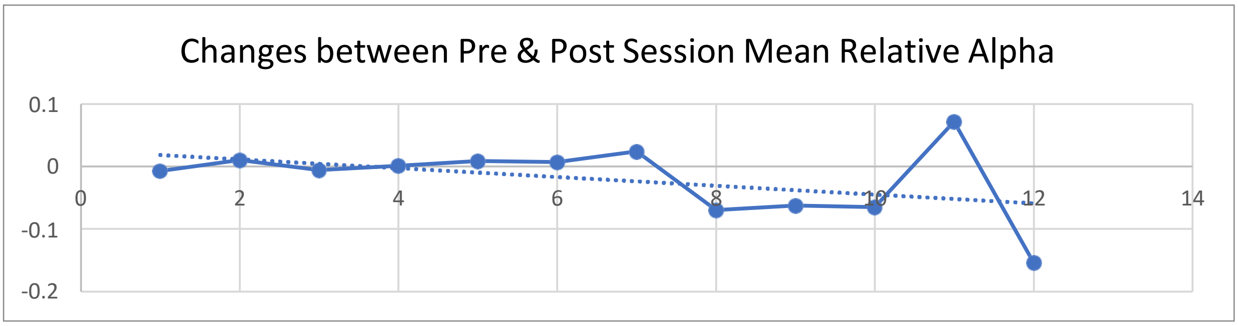
1) Slightly reduced tightness in his Right hand, seen with respect to free shoulder and wrist range of motion. More antagonist activity in pronators and wrist flexors.
2) Better in being attentive during the SynPhNe session, however, needs some improvement considering his DAR trend line and changes pre and post session in brain metrics.
3) Improvement is seen in relaxation i.e., Smiley as compared to initial sessions.
4) Reduced compensation towards left side of the body while doing the task such as picking up the pen, etc. Trunk compensation while doing the activity.

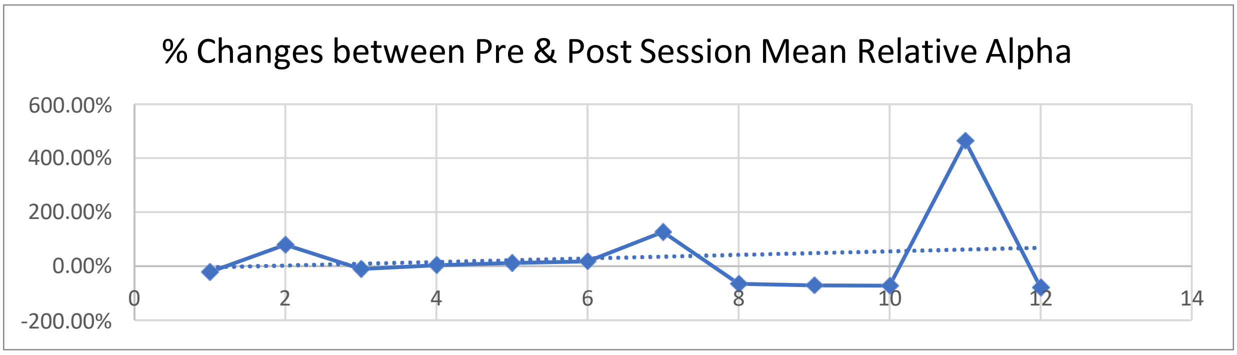
RECOMMENDATIONS:
1) Breathing exercises and trying to keep the mouth open throughout the day to facilitate more breathing. This will help in further relaxation and improving attention during the day as well as the session.
2) Stretching of right upper limb to reduce tightness of all the muscle groups. Activation of supinator, wrist extensor and finger extensor.
3) More of reaching activities to reduce the trunk compensation.
4) Trying to engage left brain throughout the day, further attaining the symmetry between left and right hemisphere.

RESTING STATE BRAIN METRICS SUMMARY

<u>Relative Alpha</u>



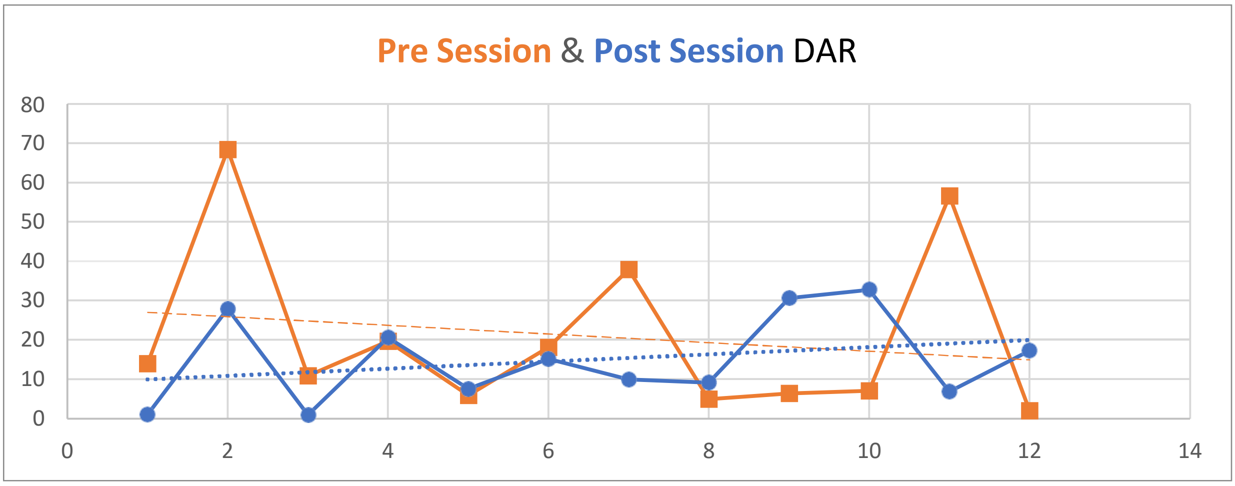


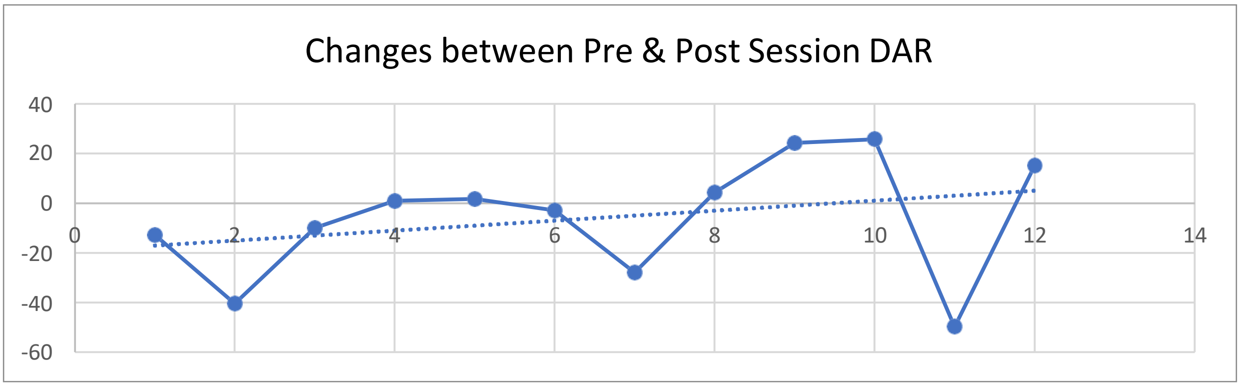


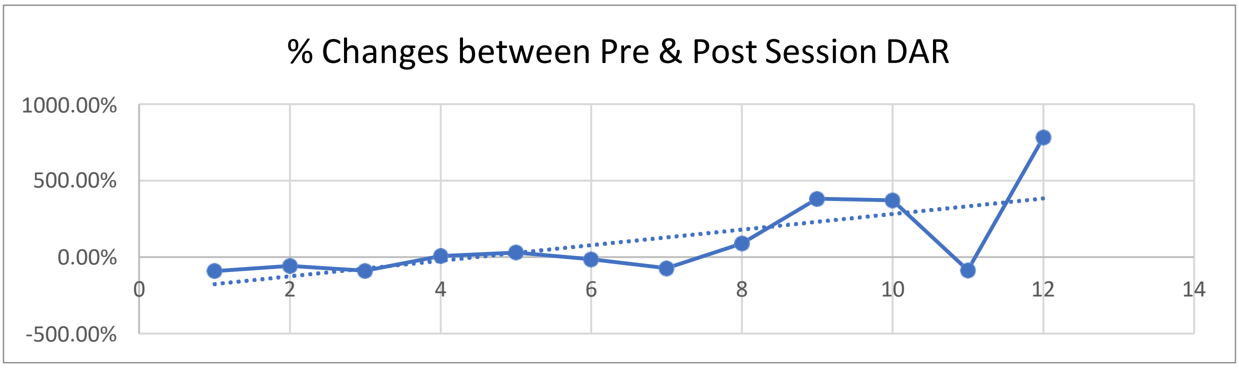
Relative Alpha is a measure of ability to remain relaxed while attempting to do the task at hand efficiently.
Ascending slope: improved ability to maintain or enhance relaxation during movement/task execution.
Descending slope: the person is tiring during the sessions and possibly, compensating.

Both Pre and Post session graphs are showing ascending slope which indicates he is showing improvement in relaxation. However, pre session trend line shows much better improvement especially after 8th session, as compared to post session trend line.

Delta to Alpha Ratio (DAR)





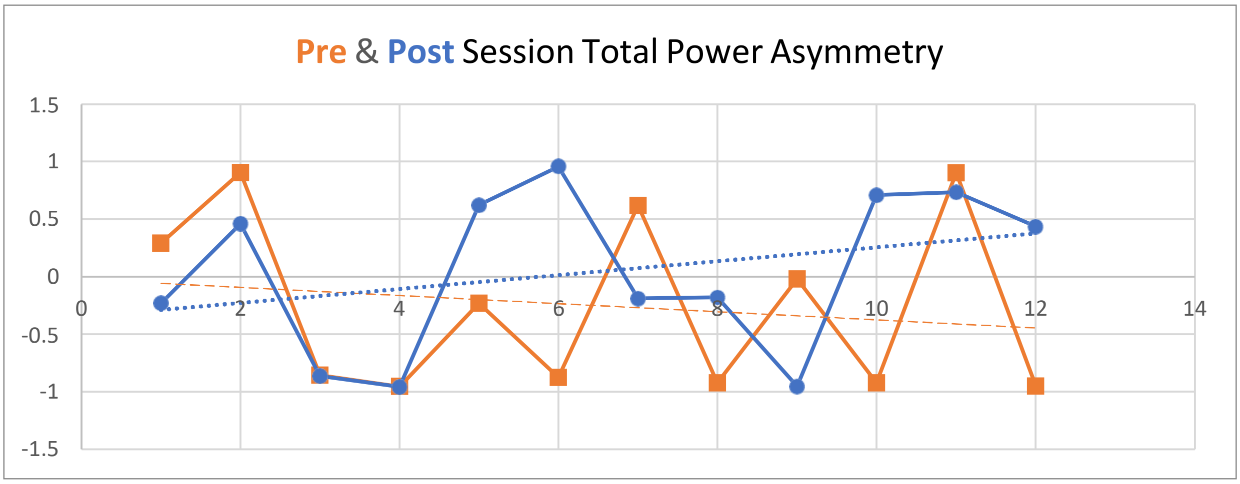


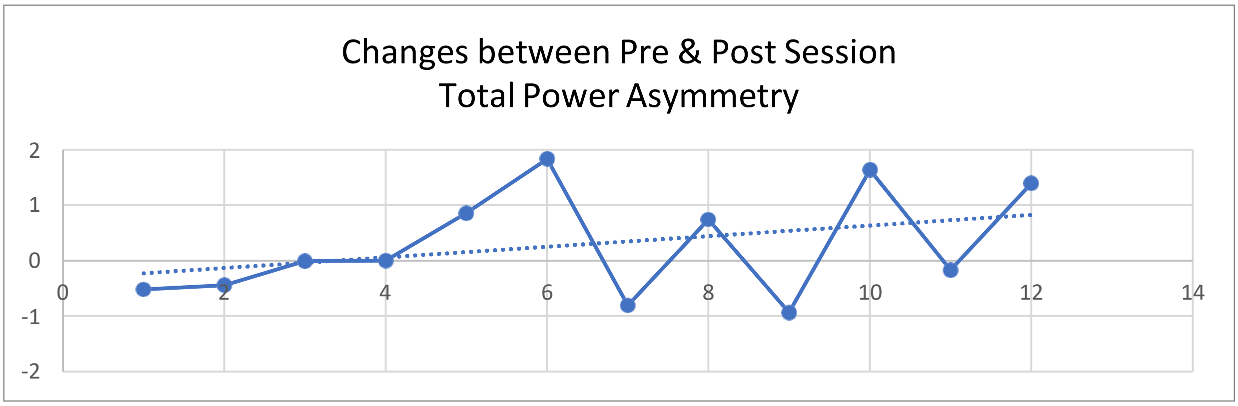
DAR is a measure of ability to remain attentive while being relaxed.
Descending slope: the person is improving attention skills and able to remain focused longer
Ascending slope: the person is experiencing attention fatigue during the session or across multiple sessions

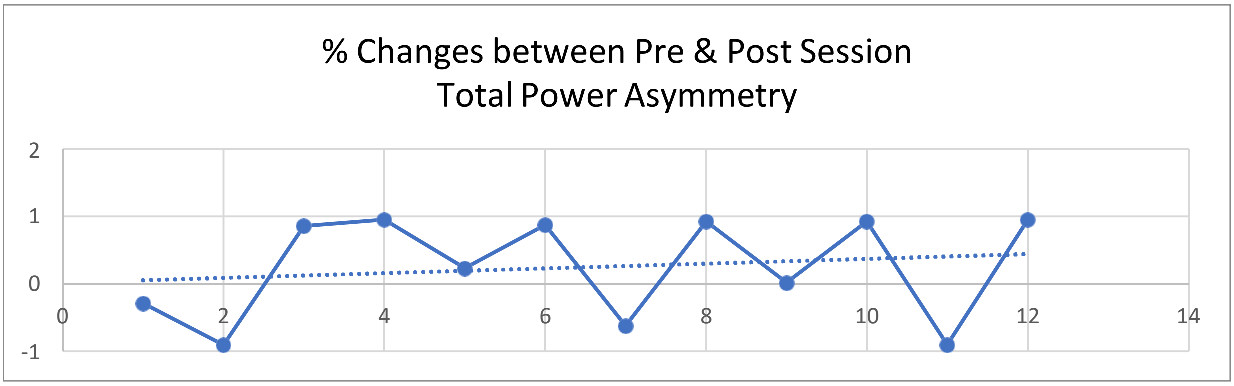
Pre session trend line shows a descending slope which indicates that he is showing improved attention during the day. But ascending slope for post session graph shows he is having difficulty maintaining the attention during the session or while performing tasks. Looking at the graphs, pre and post session changes are seen majorly in only few sessions.

Asymmetry

Total Asymmetry





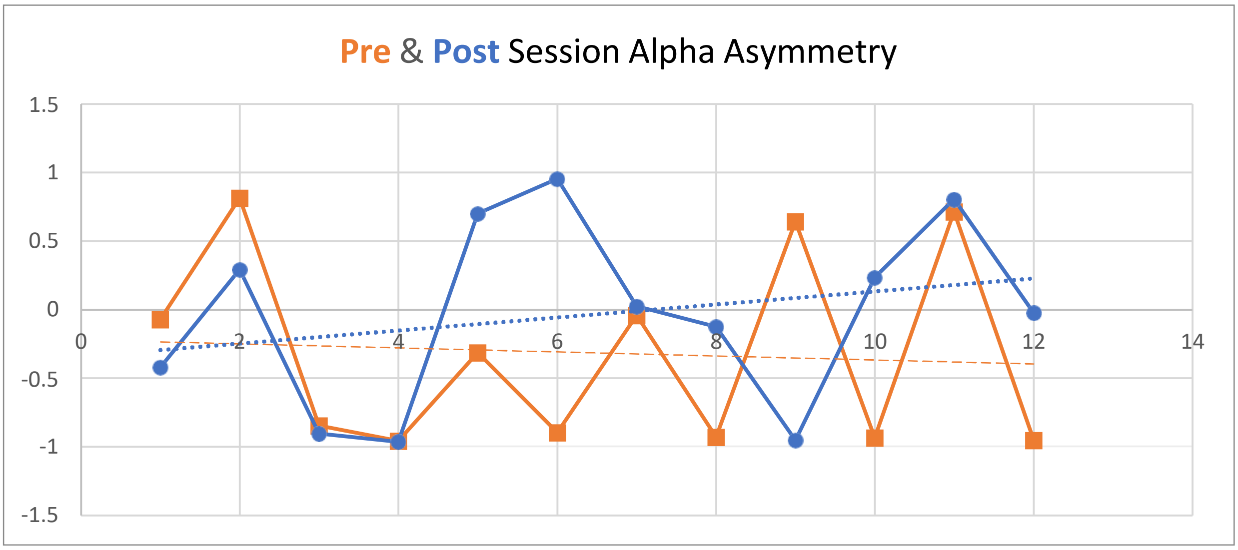


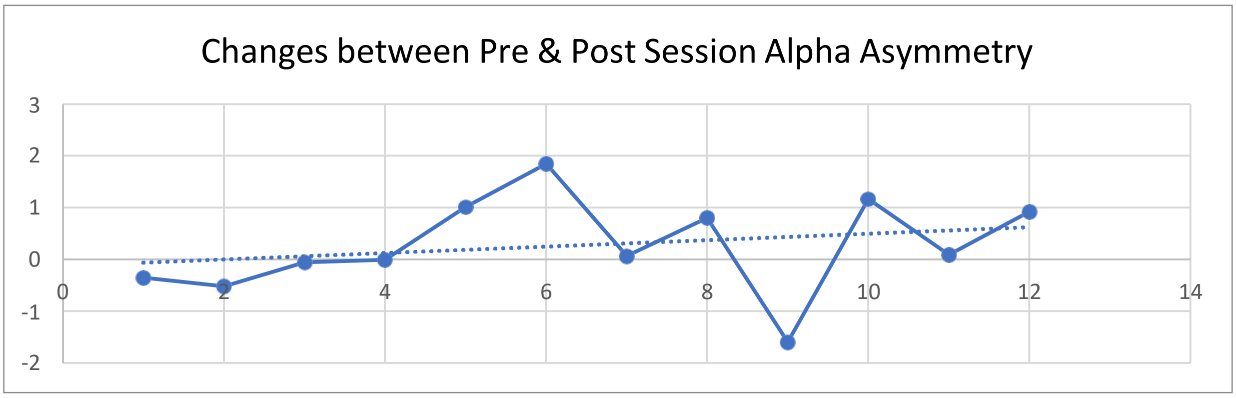
Reduced asymmetry between left and right brain hemispheres is a marker for better left-right brain cooperation in everyday tasks. Ideally, both trends' lines should be heading towards / maintaining near zero.

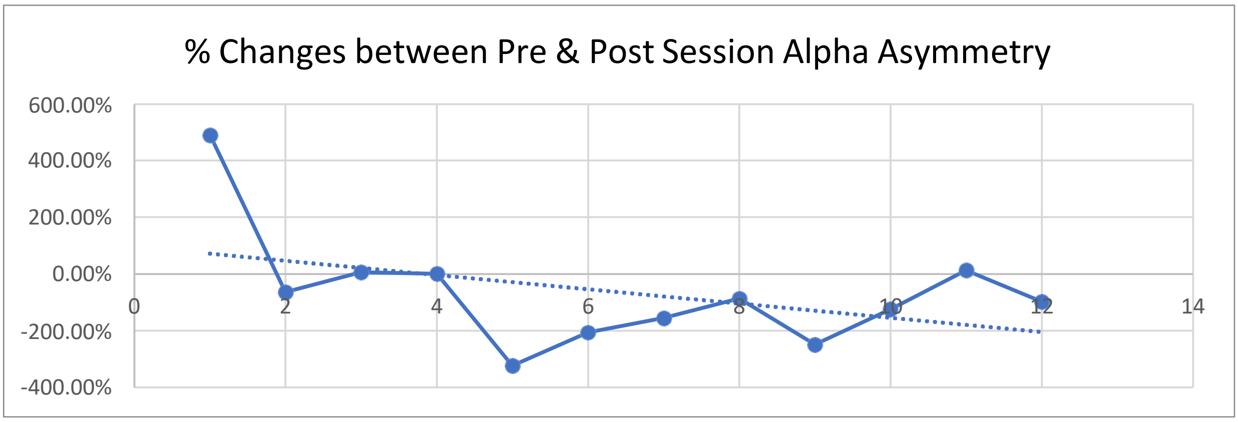
A stable resting state alpha asymmetry at zero value indicates that symmetry is restored.

Pre session trend line shows descending slope in the negative side, indicating that patient is engaging more of right hemisphere of his brain during the day. This suggests that graph is moving far from symmetry. However, in post session trend line, we can see that it is moving from negative to positive side, indicating that he is engaging his left hemisphere while performing the task with his right hand.

Alpha Asymmetry







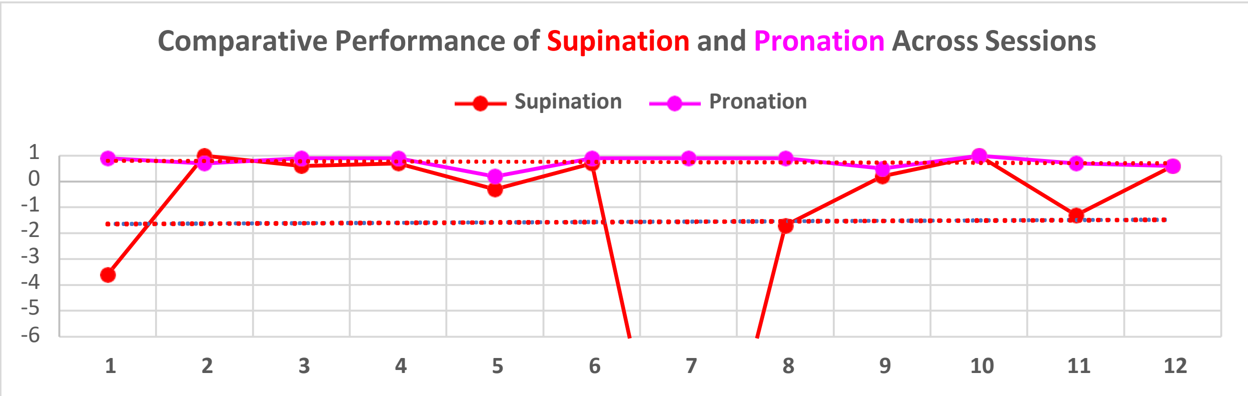
Reduced alpha asymmetry between left and right brain hemispheres is a marker for better left-right brain cooperation in everyday tasks due to both hemispheres having similar alpha activity, indicating that both hemispheres are well oxygenated. Stable values at/near zero indicates that alpha symmetry is resilient.

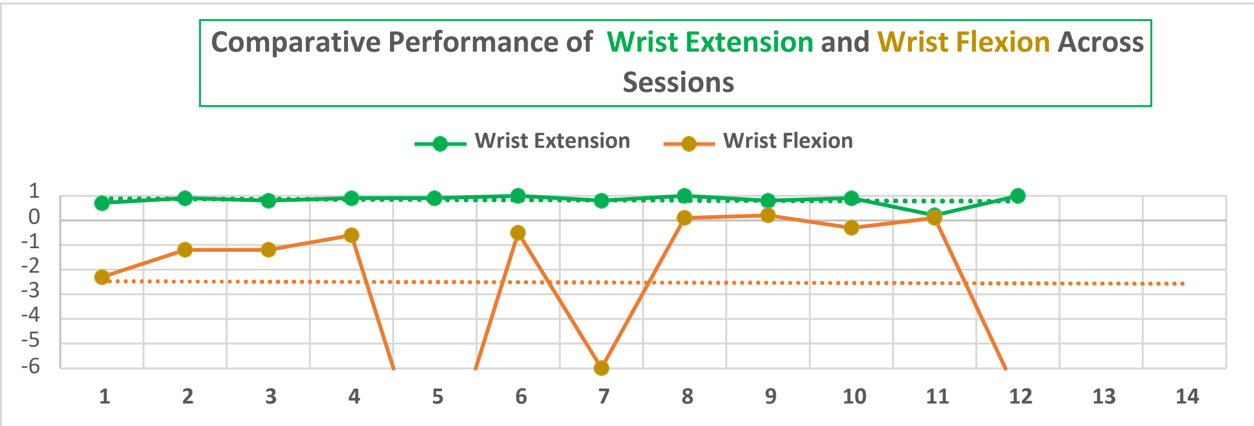
Pre session trend line shows descending slope in the negative side, indicating that right hemisphere of his brain is more oxygenated throughout the day. However, post session trend line is moving from negative to positive side, indicating that left hemisphere is getting oxygenated too, trying to attain symmetry.

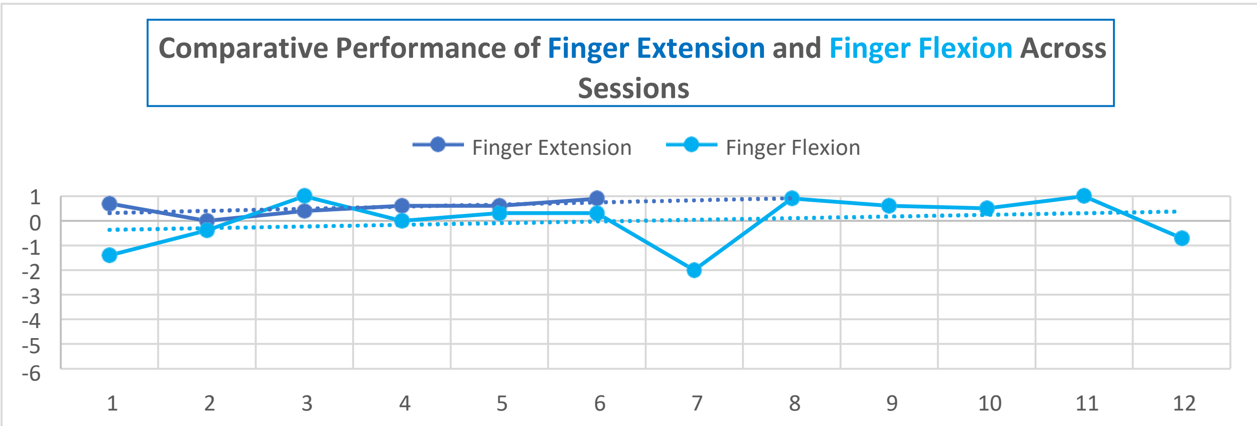
ACTIVE STATE MUSCLE METRICS SUMMARY

WHAT WE LOOK FOR:
1. Y-Axis values around 0.5 are desirable
2. Y-Axis values that are negative indicate loss of muscle control.
3. Trend lines that move towards 0 - 1 indicate improved Bt.
4. Trend lines that move away from 0 -1 indicate deterioration of Bt.

Muscle Trend





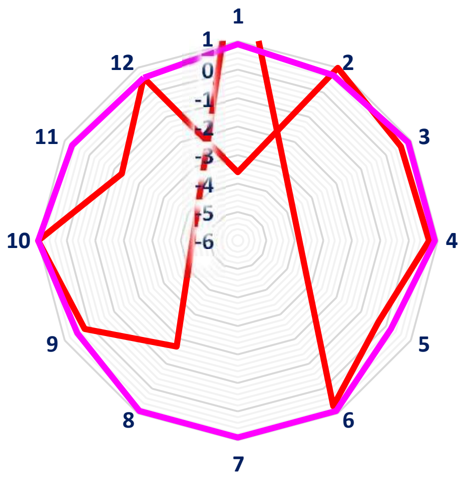


MUSCLE METRIC SUMMARY
PRONATION- muscle trend line is steady between 0 to 1, indicating a good agonist contraction through all the sessions. SUPINATION- muscle trend line is steady in the negative, indicating hyperactivity of antagonist muscle, vastly seen in 1st, 7th, 8th and 11th sessions. WRIST EXTENSION- muscle trend line is steady between 0 and 1, more towards 1, indicating a good agonist contraction through most of the sessions. WRIST FLEXION- muscle trend line is steady on the negative side, indicating hyperactivation of antagonist muscles but we can see that he is trying to use agonist muscle from 8th session. FINGER EXTENSION- muscle trend line is seen going from near 0 towards 1, indicating good agonist contraction in most of the sessions. FINGER FLEXION- muscle trend line shows graph going from negative towards positive, indicating improvement in activation of agonist muscle which can be seen in few sessions in between.

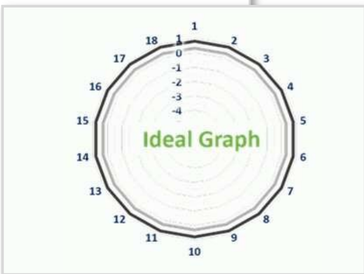
MUSCLE GROUPS COMPARATIVE REPORT

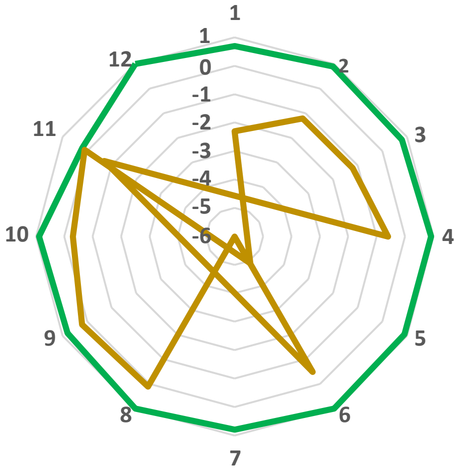
WHAT WE LOOK FOR:

1. Y-Axis values Between 0 and 1 are desirable
2. Y-Axis values that are negative indicate loss of muscle control.

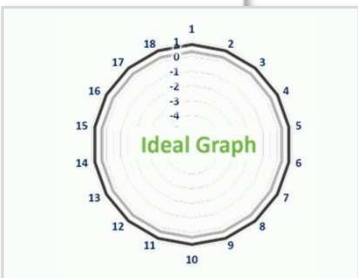


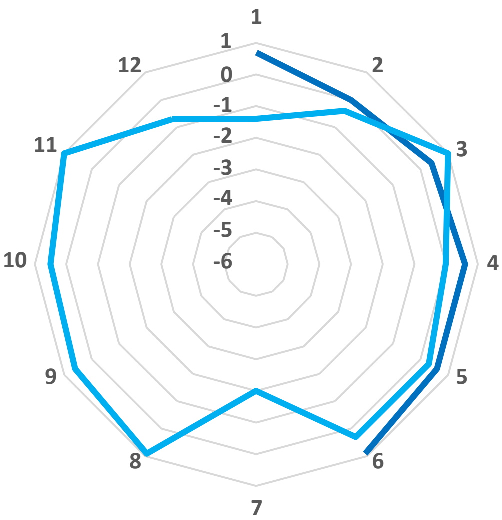
Performance of
Supination and **Pronation**
Across the Sessions





Performance of
Wrist Extension and
Wrist Flexion
Across the Sessions





Performance of
Finger Extension and
Finger Flexion
Across the Sessions